

Module 3, Lesson 3

AI 101 FOR MECHANICAL ENGINEERS

Available inputs and causal claims

Presentation Structure

1. Input Availability and Data Chronology

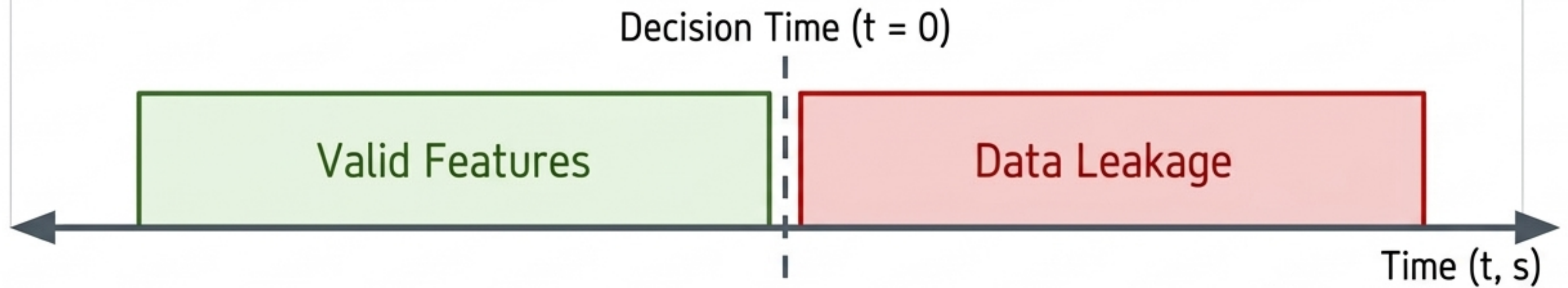
2. Statistical Association versus Causation

3. Engineering Application and Assessment

1. Input Availability and Data Chronology

Establish temporal constraints for predictive engineering models.

Temporal Limits of Predictive Data



Valid predictors require data accessible at decision time.

Future measurements leak into historical training data.

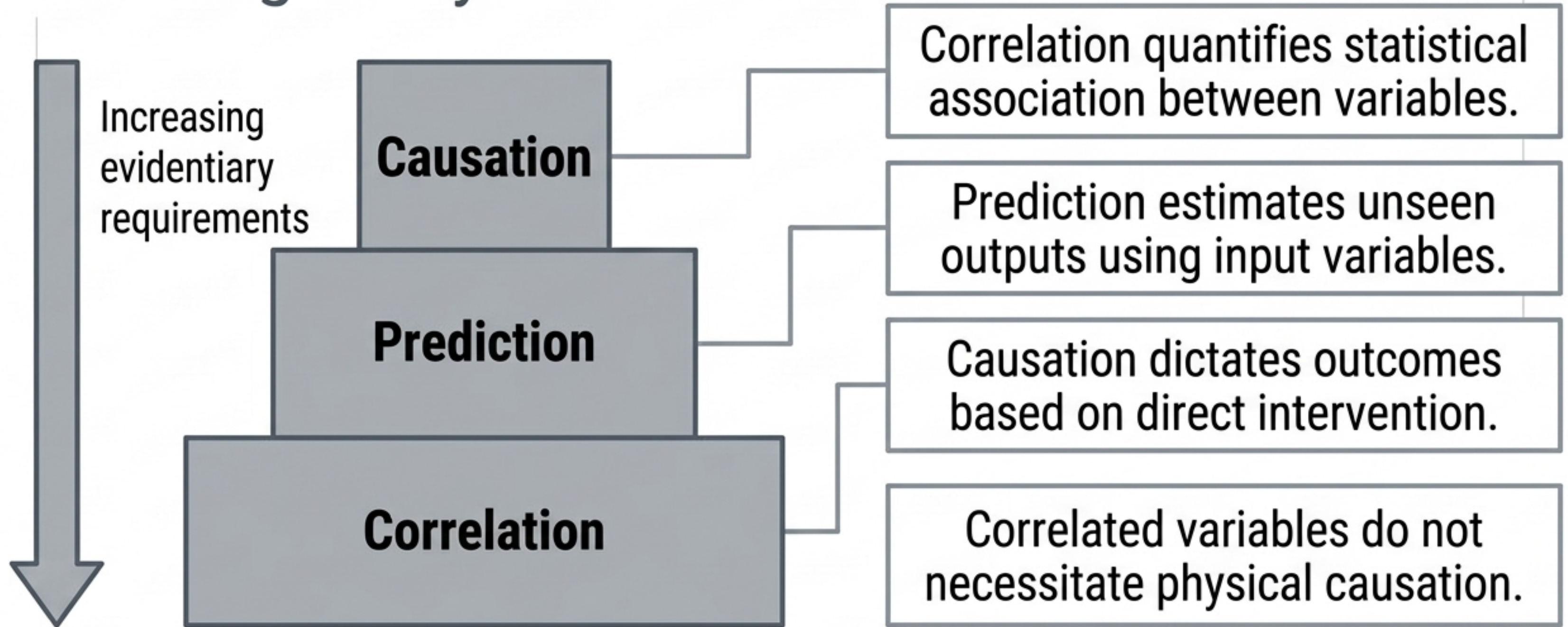
Post-failure repair codes violate temporal precedence.

Signal-window features must exclude future measurements.

2. Statistical Association versus Causation

Differentiate statistical correlation, predictive capacity, and direct physical intervention.

Defining Analytical Claims



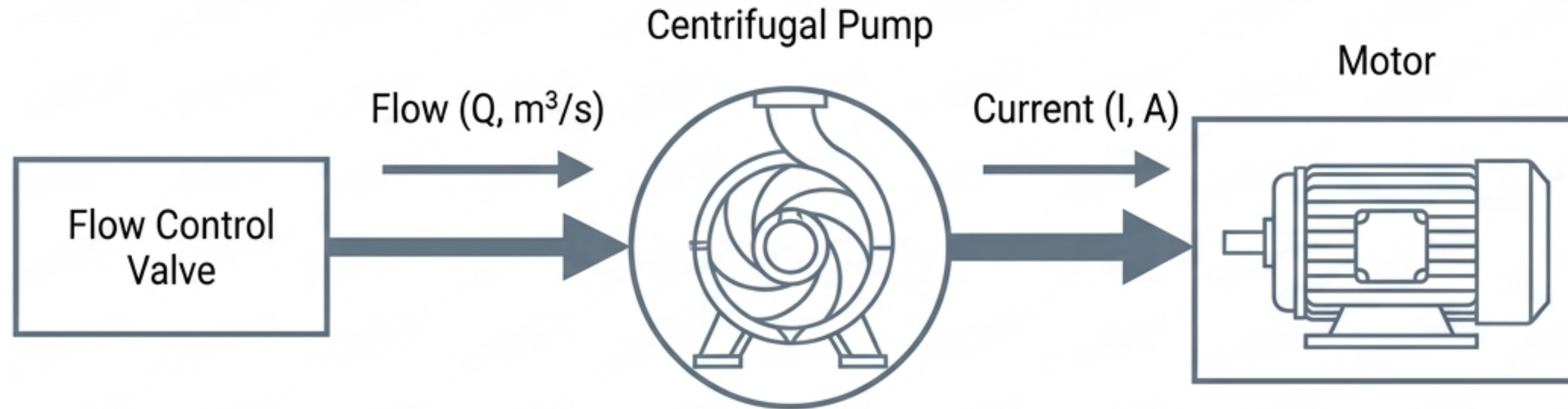
Framework for Analytical Claims

Claim Type	Question Answered	Required Evidence
Correlation	Do variables vary together?	Observational data
Prediction	Do inputs estimate unseen outputs?	Unseen validation data
Causation	Does an intervention change outcomes?	Controlled experiment
Leakage Check	Does this input exist at decision time?	Chronological verification

3. Engineering Application and Assessment

Apply causal and temporal frameworks to physical mechanical systems.

Flow and Motor Current Relationship



Records demonstrate flow and motor current increase together.

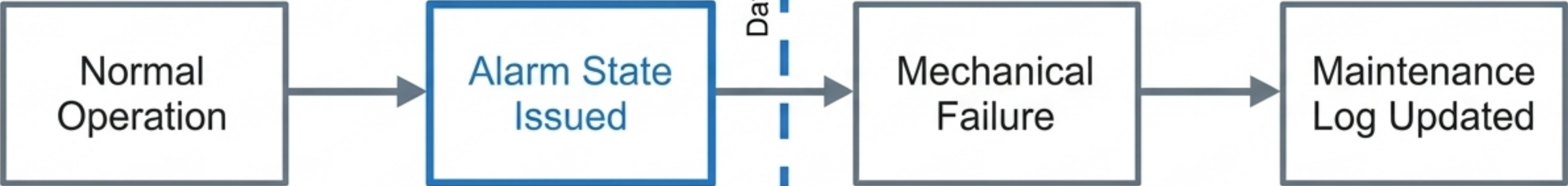
This establishes statistical association, not physical causation.

Causation relies on pump physics and control logic.

Predictive coefficients alone cannot establish causal relationships.

Identifying Temporal Data Leakage

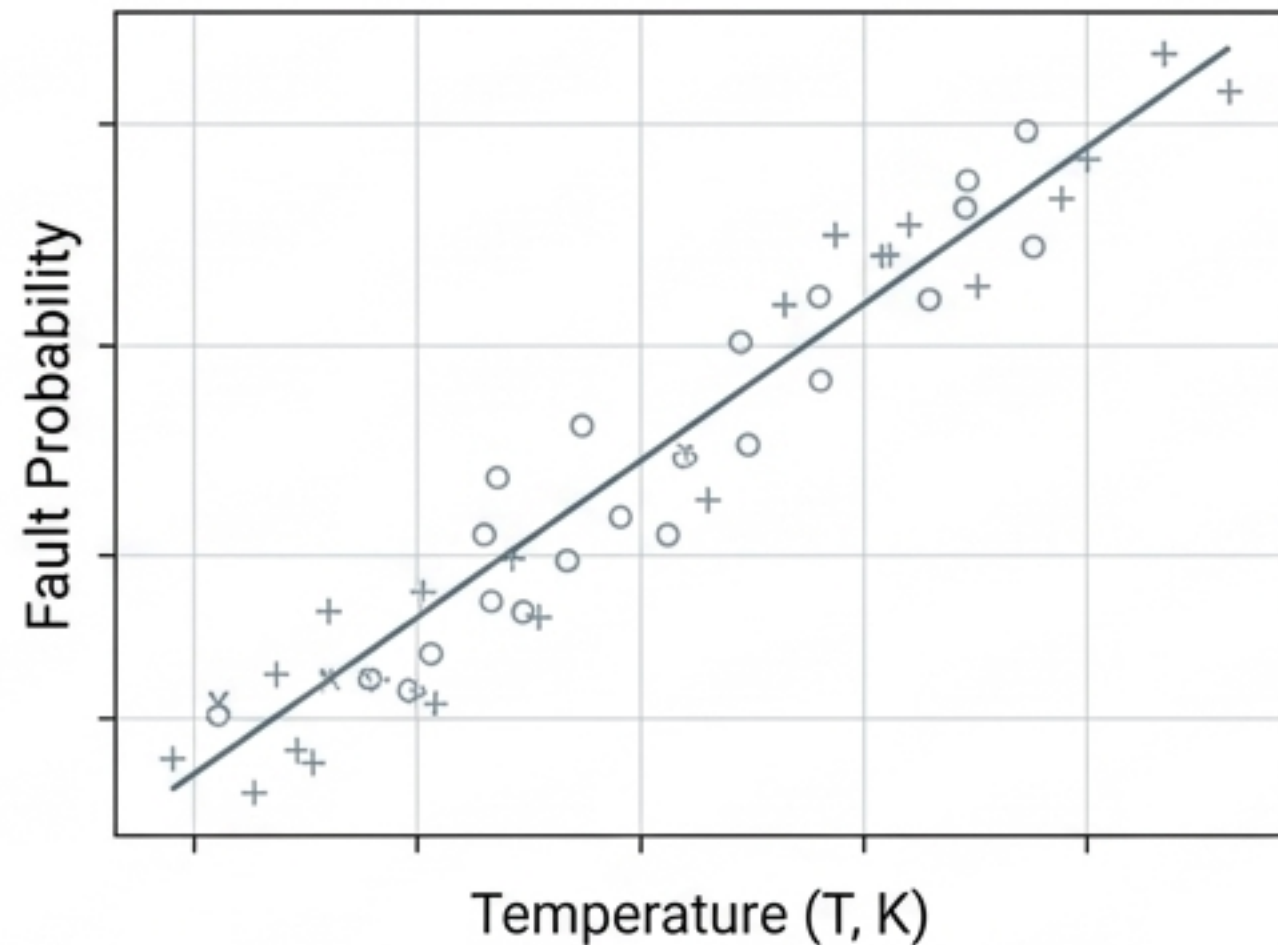
Data Collection Boundary



- Assess inputs for temporal validity before alarm issuance.
- Current vibration RMS remains chronologically valid.
- Repair codes constitute future information leakage.
- Predictions must utilize strictly antecedent operational values.

Limits of Predictive Associations

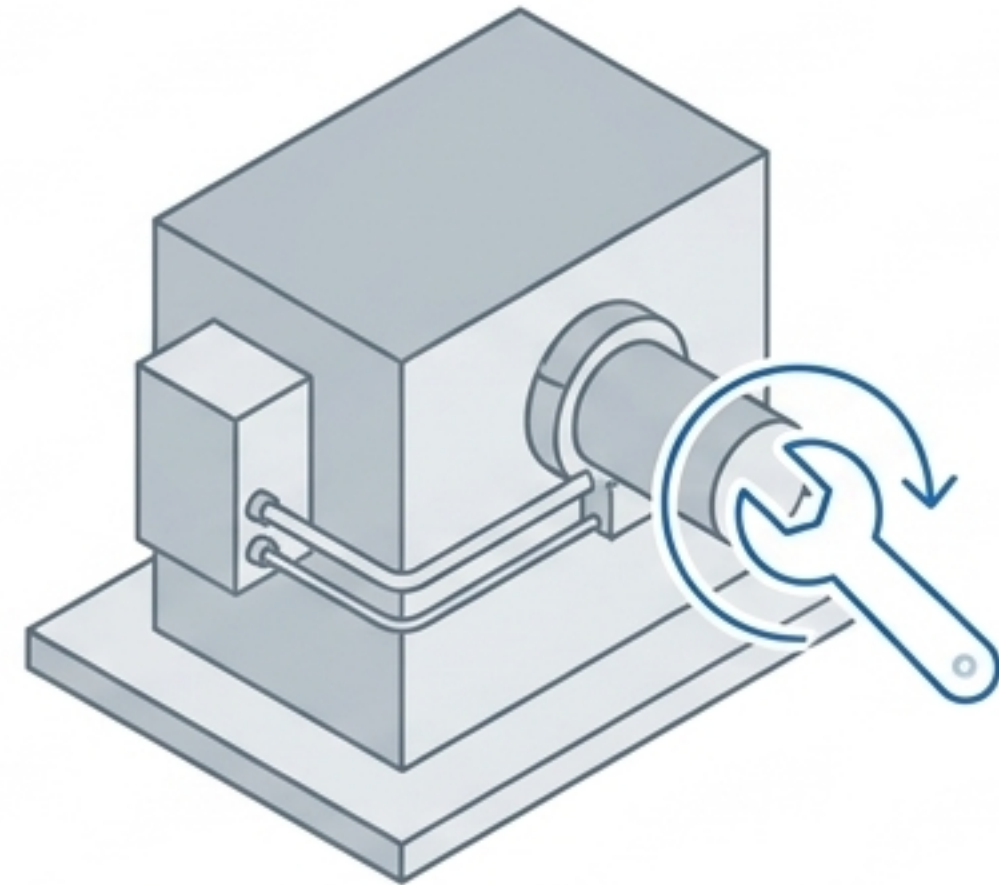
Statistical Association



Predictive models identify associations without causal proof.

Prediction targets estimates; causation dictates physical reality.

Physical Intervention (Ineffective)



Changing a correlated sensor reading prevents no failures.

Physical interventions require distinct empirical validation.

Key Points

1. Valid models strictly require data chronologically preceding the operational decision.

2. Statistical correlation and prediction differ fundamentally from physical causal intervention.

3. Predictive associations in mechanical systems demand independent physical and experimental verification.